

# Learning and Efficiency in the Market for Physician Referrals

## SUPPLEMENTAL APPENDIX

### PRELIMINARY

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## 1 Descriptive Statistics

This section provides additional details of our analytic data and subsequent referral patterns.

### 1.1 PCP Identification

Our method identifies a patient’s PCP using a carrier-claims lookback window, selecting the most-visited PCP in pre-surgery E&M claims. Table 1 compares this approach with the referring physician listed on the surgeon’s claim (`rfr_physn_npi`). The listed referring physician field is not populated in roughly 32% of cases, and when it is populated, it most often lists the surgeon (self-referral, about 43%) or a non-PCP specialist (about 7%). Only about 18% of cases list a PCP as the referring physician. Among cases where both methods identify a PCP, concordance exceeds 82% in every year.

Figure 1 shows the composition of the listed referring physician field alongside our lookback coverage rate. Our lookback method identifies a PCP for approximately 80% of procedures, providing substantially better coverage than the listed field alone.

Table 1: **PCP Identification Method Validation**<sup>a</sup>

| Year | Procedures | Composition of Referral Field (%) |               |         |      | Lookback     |                 |
|------|------------|-----------------------------------|---------------|---------|------|--------------|-----------------|
|      |            | Not Pop.                          | Self-Referral | Non-PCP | PCP  | Coverage (%) | Concordance (%) |
| 2013 | 445,274    | 30.1                              | 41.6          | 7.5     | 20.8 | 81.5         | 82.8            |
| 2014 | 456,377    | 31.2                              | 41.6          | 7.1     | 20.1 | 80.9         | 82.7            |
| 2015 | 502,927    | 31.1                              | 42.9          | 7.1     | 18.9 | 80.5         | 82.8            |
| 2016 | 542,126    | 31.5                              | 43.7          | 7.1     | 17.8 | 79.8         | 82.4            |
| 2017 | 557,277    | 32.6                              | 44.1          | 6.7     | 16.6 | 79.0         | 82.7            |
| 2018 | 506,138    | 34.4                              | 43.5          | 6.5     | 15.6 | 78.2         | 82.1            |
| All  | 3,010,119  | 31.9                              | 43.0          | 7.0     | 18.2 | 79.9         | 82.6            |

<sup>a</sup>Comparison of our carrier-lookback PCP identification method with the referring physician listed on the surgeon’s claim (`rfr_physn_npi`). “Not Pop.” indicates the referral field is not populated. “Self-Referral” indicates the listed referring physician is the operating surgeon. “Non-PCP” indicates a non-PCP specialist. “PCP” indicates a primary care physician. “Lookback Coverage” is the share of procedures for which our method identifies a PCP. “Concordance” is the share of cases where both methods identify a PCP and they agree.

## 2 Identification

We formally show identification of the myopic model, and then we discuss identification of the dynamic model.

### 2.1 Myopic Model

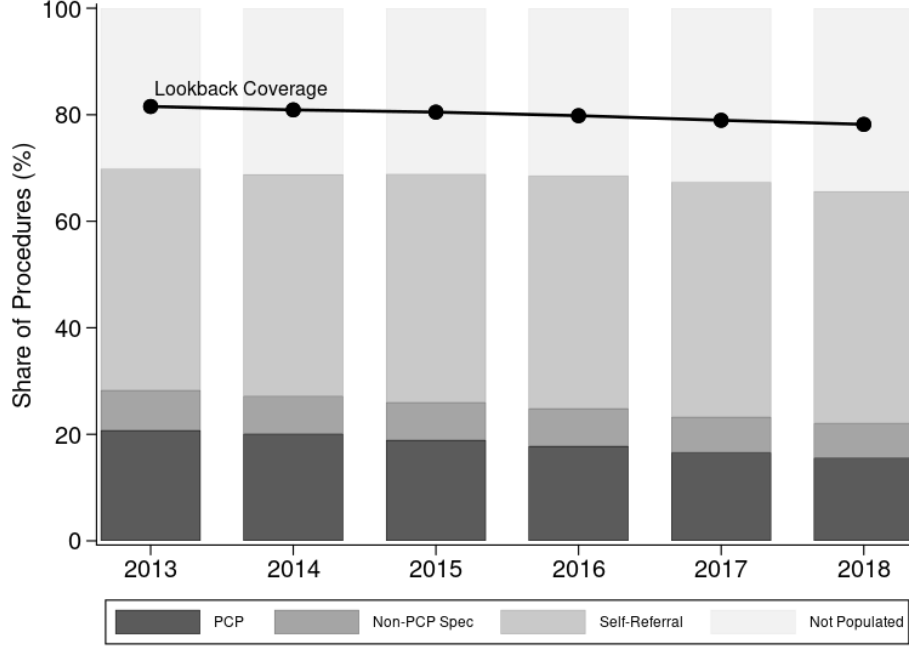
Given a distribution for the shocks  $\epsilon$ , the conditional choice probabilities identify differences in utility between the alternatives. The effects of the pairwise characteristics,  $u(x)$ , and the specialist fixed effects,  $\xi$ , are identified subject to standard normalizations (e.g., one fixed effect is set equal to zero). Because the number of patients at a specialist,  $n_{jt}$ , is endogenous, the congestion effect,  $c(n_{jt}, z_j)$ , is identified with an instrument for the probability that patients *other than patient t* see specialist  $j$ . A natural instrument for this is the distances between *other* patients and specialist  $j$  (see Richards-Shubik et al., 2022).

This leaves the terms  $\alpha m$  and  $f(e)$ . The parameter  $\alpha$  is identified in the limit as the number of past patients sent to a specialist grows large (i.e., so-called “identification at infinity”). Recall  $e_{ijt} \equiv \sum_{s=1}^{t-1} D_{ijs}$  and let  $\bar{y}_{ijt} \equiv \sum_{s=1}^{t-1} Y_{ijs}/e_{ijt}$ . Then,

$$\alpha m_{ijt} = \alpha \frac{a_0 + \sum_{s=1}^{t-1} Y_{ijs}}{a_0 + b_0 + \sum_{s=1}^{t-1} D_{ijs}} = \alpha \frac{a_0/e_{ijt} + \bar{y}_{ijt}}{(a_0 + b_0)/e_{ijt} + 1}.$$

Hence,  $\lim_{e_{ijt} \rightarrow \infty} \alpha m_{ijt} = \alpha \bar{y}_{ijt}$ . Intuitively,  $\alpha$  is identified by variation in the success rate

Figure 1: Composition of the listed referring physician field and carrier-lookback coverage over time.<sup>b</sup>



<sup>a</sup>Stacked bars show the composition of the `rfr_physn_npi` field on the surgeon's claim. The line shows the share of procedures for which our carrier-lookback method identifies a PCP.

<sup>b</sup>Stacked bars show the composition of the `rfr_physn_npi` field on the surgeon's claim. The line shows the share of procedures for which our carrier-lookback method identifies a PCP.

among specialists to whom a PCP has sent many patients in the past.

Next, the strength of the priors,  $(a_0 + b_0)$ , is identified by how the marginal effect of the success rate changes with the number of past patients. Rearranging  $\alpha m_{ijt}$  differently, we have

$$\alpha m_{ijt} = \alpha \frac{a_0}{(a_0 + b_0) + e_{ijt}} + \alpha \frac{e_{ijt}}{(a_0 + b_0) + e_{ijt}} \times \bar{y}_{ijt}. \quad (1)$$

With  $\alpha$  identified, the second term in equation (1) has only one unknown parameter,  $\eta \equiv (a_0 + b_0)$ . The marginal effect of  $\bar{y}$  increases with  $e$ , and  $\eta$  governs that interaction.

To identify the prior mean,  $\rho \equiv \frac{a_0}{a_0 + b_0}$ , we use the observed average success rate in the market. This assumes that PCPs have a general awareness of the overall success rate of the procedure in their market. It might instead be possible to identify  $a_0$  from the choice data, but this would require a strong functional form assumption to separate the term  $\alpha \frac{a_0}{(a_0 + b_0) + e}$  from the familiarity effect  $f(e)$ .

Finally, then, the familiarity effect is identified by the remainder of the marginal effect of experience with a specialist that is not absorbed by  $\alpha m$  (i.e., what is left over from  $\alpha \frac{a_0}{(a_0+b_0)+e}$ ). A standard normalization is required, such as  $f(0) = 0$ . Also  $f$  must be bounded as  $e \rightarrow \infty$ ; otherwise, familiarity would dominate the effects of all other attributes as  $e \rightarrow \infty$ . This does not seem desirable from a modeling perspective, and it would nullify the “identification at infinity” argument for  $\alpha$ .

We start by proving the identification of the myopic model. To review (and simplify), the utility to PCP  $i$  for referring patient  $t$  to specialist  $j$  is

$$U_{ijt} \equiv \alpha m_{ijt} + f(e_{ijt}) + \xi_j + \epsilon_{ijt},$$

where  $m_{ijt}$  is the current mean of the beliefs about the probability of success at specialist  $j$  (using the history before patient  $t$ ),  $e_{ijt}$  is the amount of experience with specialist  $j$  (the number of past patients  $i$  sent to  $j$ ),  $f(e_{ijt})$  is the subjective utility from familiarity with specialist  $j$ ,  $\xi_j$  is an unobservable specialist fixed effect, and  $\epsilon_{ijt}$  is an idiosyncratic shock with a known distribution. Also let  $\check{y}_{ijt}$  denote the number of successes among past patients that  $i$  sent to  $j$ .

Given a parametric assumption on the error term,  $\epsilon$ , the observable choice probabilities directly identify differences in utility, excluding the error term: e.g.,  $(U_{ijt} - \epsilon_{ijt}) - (U_{ikt} - \epsilon_{ikt})$ . To simplify these expressions, let  $v$  denote utility without the error term:  $v \equiv U - \epsilon$ .

To be somewhat more precise, there is a finite number of specialists but large numbers of patients and PCPs. The choice probabilities are

$$\Pr(j | (e_{ikt}, \check{y}_{ikt})_{k=1}^J),$$

where “ $j$ ” means the event that specialist  $j$  is chosen. We may need consistent estimates of the choice probabilities for cases where  $e_{ikt} = 0, \forall k$ , and where  $e_{ikt} \rightarrow \infty, \forall k$ . Hence the number of PCPs may need to go to infinity, but the number of patients must grow faster.

We identify the remaining components of  $v$  in steps as follows.

1. Specialist fixed effects ( $\xi$ ): Any cases where the observables are the same for two specialists can be used to identify the fixed effects (up to a standard normalization;

e.g., one fixed effect is set to zero). Because if  $e, \check{y}, x, n$  are the same, then

$$v_{ijt} - v_{ikt} = \xi_j - \xi_k.$$

In particular, cases where  $e = 0$  (and hence  $\check{y} = 0$ ) are useful here.

2. Altruism parameter ( $\alpha$ ): This uses cases with large past experience, formally the limit when  $e \rightarrow \infty$ . Let  $e_{ijt} \rightarrow \infty$  and  $e_{ikt} \rightarrow \infty$  (and for convenience  $x_{ijt} = x_{ikt}$  and  $n_j = n_k$ ). Then

$$v_{ijt} - v_{ikt} = \frac{\alpha \bar{y}_{ijt}}{0 + 1} + \xi_j - \frac{\alpha \bar{y}_{ikt}}{0 + 1} - \xi_k.$$

Hence

$$\alpha = \frac{(v_{ijt} - \xi_j) - (v_{ikt} - \xi_k)}{\bar{y}_{ijt} - \bar{y}_{ikt}}$$

in these limits. Thus cases where a PCP has large past experience with two specialists, but different success rates with them, identify the altruism parameter.

3. Strength of priors ( $a_0 + b_0$ ): The responsiveness to success with finite experience then identifies the strength of priors (i.e., the stickiness of beliefs). Let  $e_{ijt} = e_{ikt} = e$  (and for convenience  $x_{ijt} = x_{ikt}$  and  $n_j = n_k$ ). Then

$$v_{ijt} - v_{ikt} = \frac{\alpha \check{y}_{ijt}}{a_0 + b_0 + e} + \xi_j - \frac{\alpha \check{y}_{ikt}}{a_0 + b_0 + e} - \xi_k.$$

Hence

$$(a_0 + b_0) = \frac{\alpha(\check{y}_{ijt} - \check{y}_{ikt})}{(v_{ijt} - \xi_j) - (v_{ikt} - \xi_k)} - e,$$

and all the objects on the right are observable or already identified.

4. Effect of familiarity ( $f(e)$ ): Finally, differences in experience with specialists identify the effect of familiarity. For convenience let  $x, n$  be the same, then

$$v_{ijt} - v_{ikt} = \alpha \frac{a_0 + \check{y}_{ijt}}{a_0 + b_0 + e_{ijt}} + f(e_{ijt}) + \xi_j - \alpha \frac{a_0 + \check{y}_{ikt}}{a_0 + b_0 + e_{ikt}} - f(e_{ikt}) - \xi_k.$$

Hence

$$\begin{aligned}
f(e_{ijt}) - f(e_{ikt}) &= (v_{ijt} - \xi_j) - (v_{ikt} - \xi_k) \\
&- \alpha \left[ \frac{a_0}{a_0 + b_0 + e_{ijt}} - \frac{a_0}{a_0 + b_0 + e_{ikt}} \right] \\
&- \alpha \left[ \frac{\check{y}_{ijt}}{a_0 + b_0 + e_{ijt}} - \frac{\check{y}_{ikt}}{a_0 + b_0 + e_{ikt}} \right].
\end{aligned}$$

This identifies  $f$  up to a normalization, specifically  $f(0) = 0$ .

However this also requires the value of  $a_0$ , which can be obtained if the prior ratio,  $\frac{a_0}{a_0 + b_0}$ , is calibrated separately. Intuitively, we need to fully identify the priors (both the ratio and the strength) in order to recover the effect of familiarity, because we need to know exactly how experience washes away the priors in order to separate that from the effect of familiarity.

## 2.2 Finite Sample Performance

We recover negative values of  $\alpha$  in the majority of markets, whenever  $\eta$  is set above a relatively small number. This may occur because of the way that  $\alpha$  attaches to both the number of successes and the number of patients in the model, as follows:

$$\alpha \frac{\rho\eta + y_{ijt}}{\eta + e_{ijt}}.$$

This can be split into two terms where the coefficient is constrained to be the same (because  $\rho$  and  $\eta$  are set at fixed values):

$$\alpha \frac{\rho\eta}{\eta + e_{ijt}} + \alpha \frac{y_{ijt}}{\eta + e_{ijt}}.$$

There is relatively little variation in  $y$  conditional on  $e$ , because the success rate is high. Consequently, much of the identifying variation for  $\alpha$  comes from the first term (driven by variation in  $e$ ) not the second term (driven by variation in  $y$  conditional on  $e$ ).

If PCPs are more likely to refer patients to specialists to whom they have sent more patients in the past, then  $e$  would have a positive relationship with the specified utility

function. Hence the first term, with  $e$  in the denominator, would contribute to a negative estimate of  $\alpha$ . The model includes a separate familiarity effect, specified as  $\beta \log(1 + e)$ , which should absorb this, but the distribution of the estimates of  $\alpha$  across HRRs is similar whether we include that term or not.

## 2.3 Monte Carlo Evidence on the Recovery of $\alpha$

To assess the finite-sample properties of our estimator, we conduct a Monte Carlo simulation calibrated to match key features of our data. We generate referral markets with 50 PCPs and 10 specialists, where each PCP makes 15 referrals per year. We include a burn-in period to establish referral histories and rely on a 6-year estimation window to match the estimation in our real data. Specialists are assigned heterogeneous quality levels (true failure probabilities) drawn from a normal distribution centered at the base failure rate, and each specialist receives a fixed effect drawn independently. PCPs choose specialists according to the myopic structural model presented in the main text, with beliefs updated via Bayes' rule after each patient outcome.

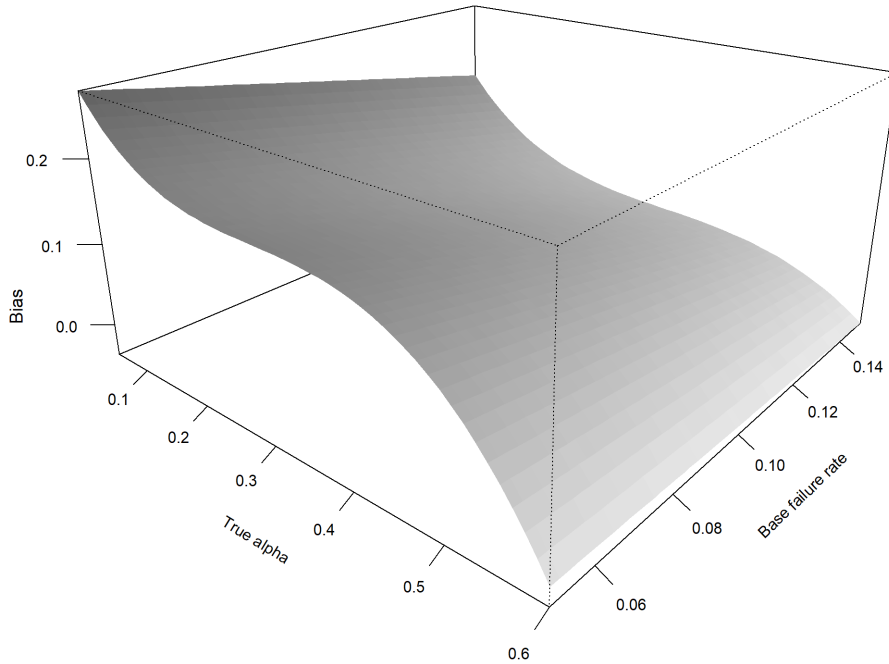
We estimate  $\hat{\alpha}$  using the same constrained maximum likelihood procedure used in the main estimation (see Section 3), with specialist fixed effects included in the conditional logit. We vary the true  $\alpha$  over the range 0.05 to 0.60, the base failure rate over 0.05 to 0.15, and the quality dispersion (standard deviation of failure rates across specialists) over 0.01 to 0.05, yielding 77 parameter configurations after excluding cases where the quality dispersion exceeds the base failure rate. For each configuration, we run 50 replications.

Figure 2 presents the bias in  $\hat{\alpha}$  as a function of the true  $\alpha$  and the base failure rate, holding quality dispersion fixed at 0.03 (approximately matching the interquartile range of failure rates in the data). The surface reveals that bias is primarily driven by the magnitude of the true  $\alpha$ , with a secondary role for the base failure rate. Bias is largest when the true  $\alpha$  is small, because the signal from patient outcomes is weak relative to the noise from the logit errors and the specialist fixed effects. Higher base failure rates improve identification by producing more frequent failures, which accelerates belief updating.

Figure 3 presents the relative bias (as a percentage of the true  $\alpha$ ) for each failure rate, again at quality dispersion of 0.03. Relative bias falls below 10% for  $\hat{\alpha} \geq 0.30$  when the

base failure rate is 0.09 or higher. Among markets with non-zero estimates of  $\hat{\alpha}$ , the mean is 0.25 for  $\eta = 1$  and 0.43 for  $\eta = 5$  in the myopic model (0.32 and 0.54 in the forward-looking model), with the 75th percentile at 0.32 and above in all cases. The overwhelming majority of markets therefore either suggest very little learning ( $\hat{\alpha} = 0$ ) or fall in the range where the estimator appears to perform well. These simulation results also support our decision to fix  $\eta$  rather than estimate it jointly with  $\alpha$ , as the likelihood surface with respect to  $\eta$  is relatively flat across the parameter configurations we examined.

Figure 2: Bias in  $\hat{\alpha}$  as a function of the true  $\alpha$  and the base failure rate, with quality dispersion fixed at 0.03. The surface is smoothed using a polynomial fit to the simulation grid. Darker shading indicates larger bias.

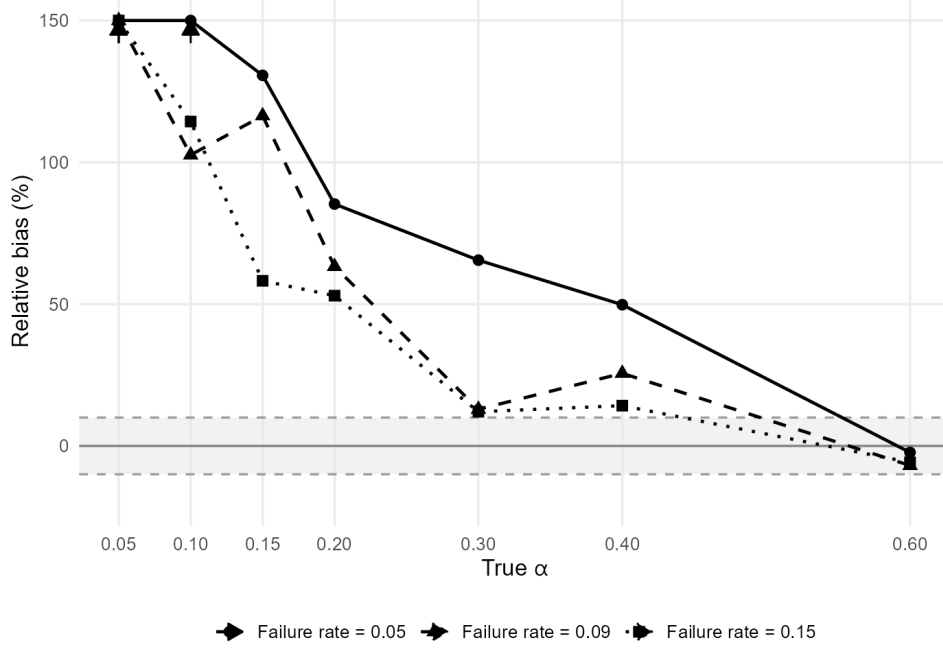


## 2.4 Simulation study on identification of $\eta$

Although we have shown that  $\eta$  is identified, our preliminary estimates of this parameter exhibited poor performance and so we used fixed values instead. Here we present a simulation study that indicates the data in our application would indeed have little power to inform an estimate of  $\eta$ .



Figure 3: Relative bias in  $\hat{\alpha}$  (as a percentage of the true value) by base failure rate, with quality dispersion fixed at 0.03. The shaded band marks  $\pm 10\%$  relative bias. Arrows indicate values exceeding 150% (truncated for readability).



We simulate a market with 2,500 patients and 25 specialists. For simplicity, the PCP-specialist histories (# referrals, # successes) are independent across patients, as though each patient were treated by a different PCP, but the PCPs have prior histories with some specialists. Each PCP has a 0.2 probability of any history with each specialist, independent across specialists. Conditional on any history with a specialist, the number of past referrals is Poisson with mean 4, and the number of past successes is binomial using the number of past referrals and the surgeon’s success probability (explained below).

The simulation model has the same specification as our empirical model, except that there is no congestion effect and the familiarity effect is specified as  $f(e_{ijt}) = \beta \ln(1 + e_{ijt})$ . The parameter values are as follows:

- $\alpha = 0.3$  (weight on outcome)
- $\pi = -0.07$  (weight on distance)
- $\beta = 0.9$  (weight on familiarity)
- $\eta = 3$  (prior strength)
- $\rho$ : explained below (prior mean)

- $\xi$ : explained below (fixed effects)

Specialist qualities,  $q_j$ , are drawn independently from a normal distribution with mean 0.9 and SD 0.04, and censored symmetrically at 0.81 and 0.99. The prior mean,  $\rho$ , is set equal to the average of these draws. The specialist fixed effects,  $\xi_j$ , are given fixed values on a grid from  $-1.5$  to  $+1.5$  with with equal intervals of  $3/25$  between values.

Figure X shows the results from four representative simulations. It plots log-likelihood values obtained using a wide range of values for  $\eta$  while keeping the other parameters at their true values. It is apparent that the likelihood function is very flat in this parameter. The log likelihood changes by only a few points over the entire span of values for  $\eta$ . Across 100 such simulations, the average range of the likelihood values is only 2.1 log points and the 95th percentile is 4.3 log points. Based on this, we do not think the data in our application would have good power to yield an informative estimate of  $\eta$ .

### 3 Estimation

To constrain  $\alpha \geq 0$  we estimate  $\ln \alpha$  and specify the beliefs term as  $\exp(\ln \alpha) \cdot m$  (or  $\exp(\ln \alpha) \cdot g(m, v)$  in the forward-looking model). We use an iterative procedure proposed in Fader et al. (1992) which makes it possible to employ standard estimation commands designed for multinomial logits where all terms are linear in parameters. The procedure is based on a first-order Taylor series approximation of the term that is nonlinear in parameters. With the exponential function this is simply

$$\exp(\ln \alpha_2) \cdot m \approx [\exp(\ln \alpha_1) + (\ln \alpha_2 - \ln \alpha_1) \exp(\ln \alpha_1)] \cdot m,$$

(because  $d \exp(\ln \alpha) / d \ln \alpha = \exp(\ln \alpha)$ ), where  $\ln \alpha_1$  and  $\ln \alpha_2$  are two candidate values.

In our case, the iteration works by setting a value of  $\ln \alpha$  (e.g.,  $\ln \alpha_1$ ) and then estimating the model with a free coefficient (e.g.,  $\hat{\pi}_1$ ) on the term  $\exp(\ln \alpha_1) \cdot m$ . Rearranging the Taylor approximation above shows how the estimated coefficient can be used to update the value of  $\ln \alpha$ , as follows:

$$\exp(\ln \alpha_2) \cdot m \approx \underbrace{[1 + (\ln \alpha_2 - \ln \alpha_1)]}_{\hat{\pi}_1} \exp(\ln \alpha_1) \cdot m;$$

hence:

$$\ln \alpha_2 = \ln \alpha_1 + \hat{\pi}_1 - 1.$$

Furthermore, because the estimated coefficient  $\hat{\pi}_1$  by definition improves the likelihood, using  $\exp(\ln \alpha_2)$  should produce a higher likelihood than  $\exp(\ln \alpha_1)$ . Fader et al. (1992) prove that this algorithm converges to the MLE of  $\ln \alpha$  under standard regularity conditions. For some intuition note that at the maximum,  $\ln \alpha_*$ , the likelihood cannot be improved, and so the value of  $\hat{\pi}$  should be one; i.e., in terms of the Taylor expansion

$$\exp(\ln \alpha_*) = \underbrace{[1 + (\ln \alpha_* - \ln \alpha_*)]}_{\hat{\pi}_*} \exp(\ln \alpha_*) = \underbrace{[1]}_{\hat{\pi}_*} \exp(\ln \alpha_*)$$

The standard error then follows from Fader et al. (1992) as

$$SE(\hat{\alpha}) = \exp(\ln \alpha_*) SE(\hat{\pi}_*)$$

because  $\hat{\alpha} = \exp(\ln \alpha_*) = \exp(\ln \alpha_* + \hat{\pi}_* - 1)$  and asymptotically we only need to consider the variation in  $\hat{\pi}_*$  (see Lemma 2 in Fader et al., 1992).

## 4 First-Stage Diagnostics for Congestion

The congestion parameter  $\gamma$  is estimated via two-stage least squares, regressing specialist fixed effects on total patient volume and HRR fixed effects, with distance-based predicted patient counts as the excluded instrument. Table 2 reports the pooled first-stage results. The first-stage coefficient is strongly significant, and the F-statistic on the excluded instrument exceeds conventional thresholds for both values of  $\eta$ .

Figure 4 presents partial scatter plots of actual versus predicted patient counts after residualizing on HRR fixed effects and a time-period indicator, confirming a strong positive relationship between the instrument and endogenous patient volume. Figure 5 further shows the distribution of HRR-level first-stage F-statistics, weighted by patient volume.

Table 2: **First-Stage Results for Congestion 2SLS**<sup>a</sup>

|            | Coefficient | SE      | F-statistic | N      |
|------------|-------------|---------|-------------|--------|
| $\eta = 1$ | 1.418       | (0.040) | 1236.5      | 36,509 |
| $\eta = 5$ | 1.415       | (0.041) | 1171.9      | 36,543 |

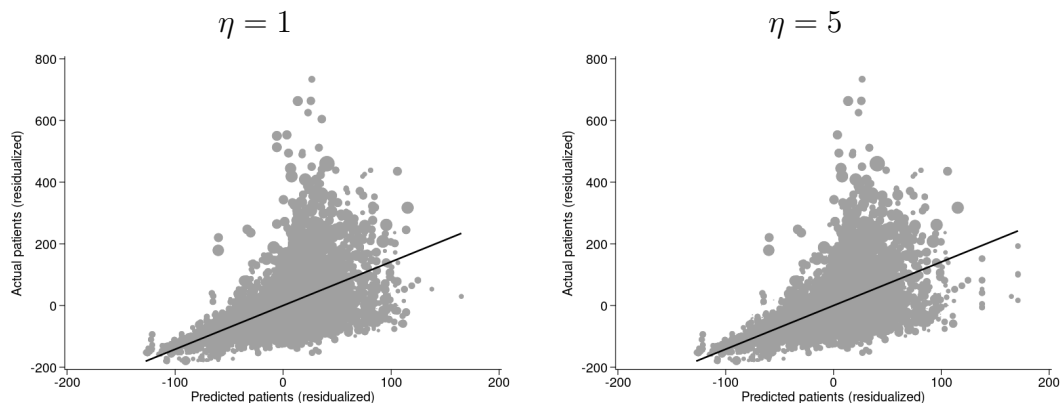
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<sup>a</sup>Pooled first-stage regression of specialist total patients on distance-predicted patient counts, with HRR fixed effects and a time-period indicator. Observations weighted by the inverse variance of the specialist fixed effect estimate. Standard errors clustered at the specialist level. Results shown for the myopic model; the forward-looking model yields nearly identical first-stage estimates because the instrument and endogenous variable do not depend on the learning model.

## 5 Counterfactuals without Familiarity

The purpose of this counterfactual is to estimate the potential changes in referrals and subsequently patient health if PCPs could rely entirely on specialist quality, with full information, and without any attachment due to prior relationships. These results are presented in Figure 8. Here, we estimate larger effects on referral probabilities, with reallocation of 35.46 percent on average across HRRs. However, the health effects of this scenario are negative on average, with a change in failure rates of -0.27 percentage points. Without any familiarity to anchor referral patterns, and with the estimated weight on quality small relative to distance preferences and specialist fixed effects, the reallocation proceeds largely independently of quality. This result reinforces the finding in the main text that existing referral relationships encode quality information, and that removing them eliminates an important source of quality sorting in the referral market.

Figure 4: Partial scatter plots of actual versus predicted patient counts, residualized on HRR fixed effects and a time-period indicator.<sup>b</sup>



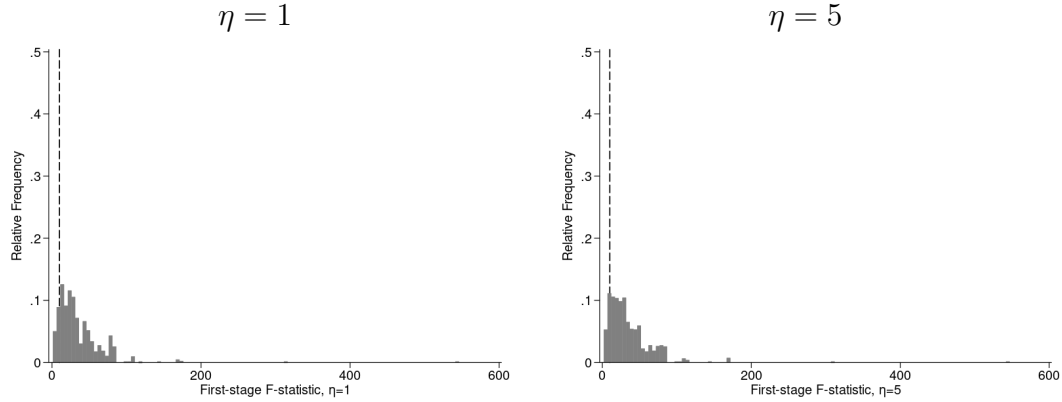
<sup>a</sup>Each point represents a specialist-time period observation, weighted by the inverse variance of the specialist fixed effect. The fitted line shows the first-stage relationship. Left panel shows  $\eta = 1$ , right panel shows  $\eta = 5$ .

<sup>b</sup>Each point represents a specialist-time period observation, weighted by the inverse variance of the specialist fixed effect. The fitted line shows the first-stage relationship. Left panel shows  $\eta = 1$ , right panel shows  $\eta = 5$ .

## References

- Cengiz, Doruk, Arindrajit Dube, Attila Lindner, and Ben Zipperer, “The Effect of Minimum Wages on Low-Wage Jobs,” *The Quarterly Journal of Economics*, August 2019, 134 (3), 1405–1454.
- Fader, Peter S., James M. Lattin, and John D. C. Little, “Estimating Nonlinear Parameters in the Multinomial Logit Model,” *Marketing Science*, November 1992, 11 (4), 372–385.
- Richards-Shubik, Seth, Mark S. Roberts, and Julie M. Donohue, “Measuring quality effects in equilibrium,” *Journal of Health Economics*, May 2022, 83, 102616.

Figure 5: Distribution of HRR-level first-stage F-statistics, weighted by patient volume.<sup>b</sup>



<sup>a</sup>Dashed line marks the conventional threshold of  $F = 10$ . Each HRR-level regression includes the distance-based instrument, a time-period indicator, and is weighted by the inverse variance of the specialist fixed effect. Left panel shows  $\eta = 1$ , right panel shows  $\eta = 5$ .

<sup>b</sup>Dashed line marks the conventional threshold of  $F = 10$ . Each HRR-level regression includes the distance-based instrument, a time-period indicator, and is weighted by the inverse variance of the specialist fixed effect. Left panel shows  $\eta = 1$ , right panel shows  $\eta = 5$ .

## Tables and Figures

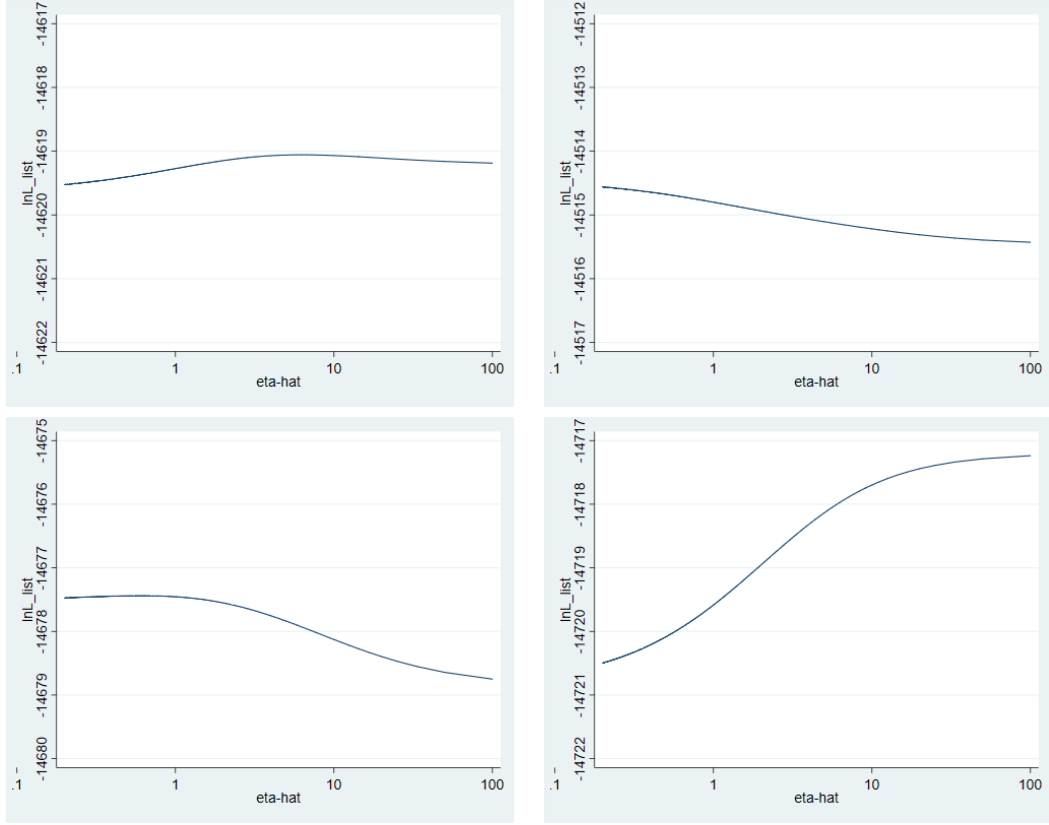
Table 3: **Sensitivity Analysis**<sup>a</sup>

| Parameter     | Mean   | Percentile |        |        |        |        |
|---------------|--------|------------|--------|--------|--------|--------|
|               |        | 10th       | 25th   | 50th   | 75th   | 90th   |
| Baseline      | 0.1924 | 0.0000     | 0.0000 | 0.1063 | 0.2766 | 0.5499 |
| Integrated    | 0.1921 | 0.0000     | 0.0000 | 0.1122 | 0.2785 | 0.5015 |
| Peer Outcomes | 0.2033 | 0.0000     | 0.0000 | 0.1137 | 0.3132 | 0.5811 |

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<sup>a</sup>Estimates of  $\alpha$  (for  $\eta = 1$ ) across alternative specifications. “Baseline” denotes our original estimates for the myopic learning model presented in the main text; “Integrated” denotes the estimates for  $\alpha$  when including a dummy variable for whether the PCP and specialist are part of the same practice; and “Peer Outcomes” denotes the estimates of  $\alpha$  when also including a measure of specialist quality based on the PCP’s peers (e.g., another measure of quality based solely on the outcomes of other PCPs’ patients in the same practice). All specifications are otherwise identical, where we also impose time-invariant congestion for computational ease.

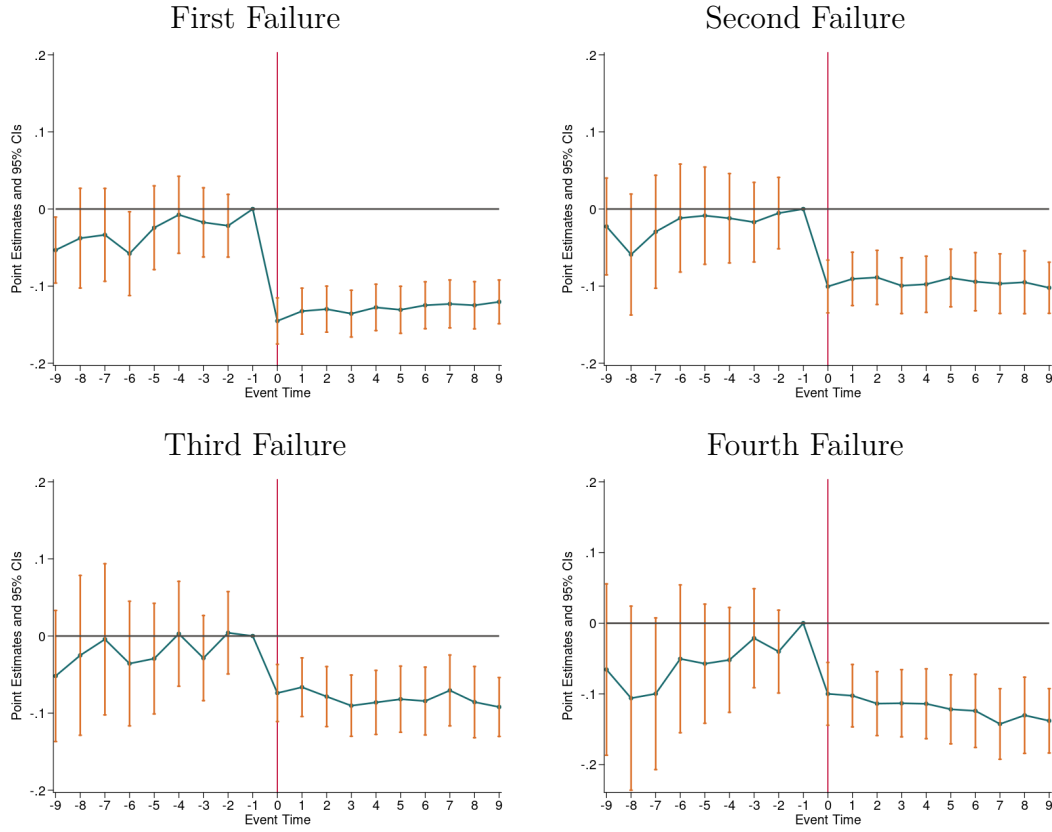
Figure 6: Log-Likelihood Values in Four Example Simulations<sup>a</sup>



<sup>a</sup>Plots of the log-likelihood (“lnL\_list”) against the candidate value of  $\eta$  (“eta-hat,” plotted on a log scale), from four example simulations. All other parameters are held at the values used to generate the simulated data, which consists of 2,500 patients and 25 specialists.

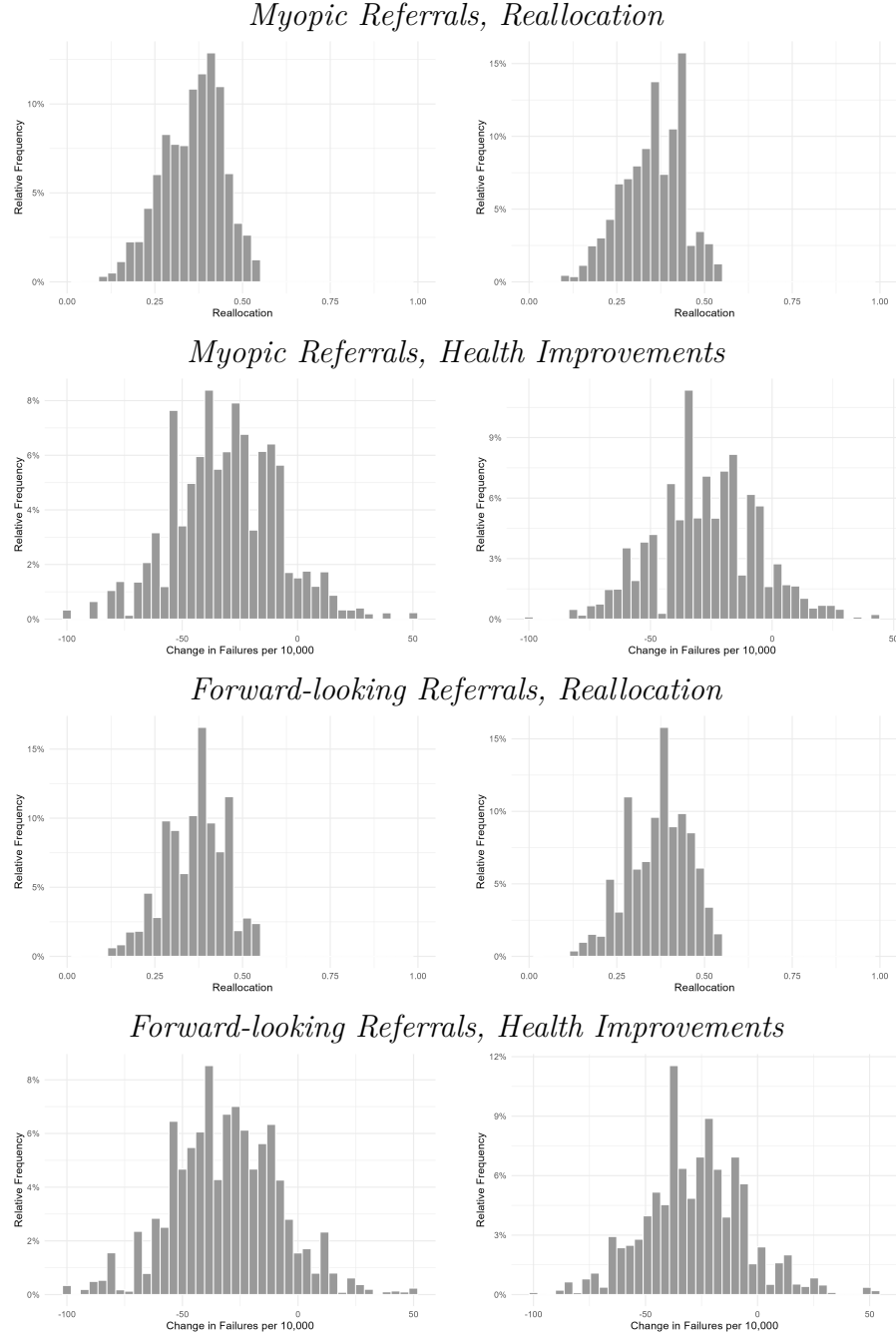


Figure 7: Event Studies by Failure Cohort<sup>a</sup>



<sup>a</sup>Figure depicts event study estimates following Cengiz et al. (2019), based on Equation 1 of the main text, estimated separately by failure cohort (e.g., first failure, second failure, etc.). Our specification includes specialist X quarter fixed effects as well as an indicator for PCP 'type'. The event study coefficient at time  $\tau = -1$  is normalized to 0.

Figure 8: Counterfactuals from Full Information and No Familiarity<sup>a</sup>



<sup>a</sup>Figure depicts the share of referrals reallocated and the resulting effects on patient health (i.e., the change in the probability of a successful surgery) under a counterfactual in which PCPs have full information about specialist quality and do not incorporate any familiarity in their referral decisions. Estimates are averaged across individuals within an HRR and presented as a histogram across HRRs. Results limited to HRRs in which our constrained maximization routine converged, 280 (277) HRRs in the myopic (forward-looking) case for  $\eta = 1$  and 281 (280) in the myopic (forward-looking) case for  $\eta = 5$ .